A Recurrent Narrow Ca II Emission Feature in the Gaseous Debris Disk of WD J2100+2122

Matias Ortiz Diez
New York University

Abstract

I identify a recurrent narrow peak in the Ca II infrared triplet of the gaseous debris disk around WD J2100+2122. Public VLT/X-shooter spectra from 2022 May 24, 2023 August 5, and 2025 May 24 show peaks near -266 km s^{-1} relative to the systemic velocity, superposed on variable broader emission. A comparison of 29 observing nights uses quiet references matched by slit width, with independent reductions and resolution checks. The measured centroid depends on the broad-profile model by up to 8 km s^{-1}. No similarly distinct peak is identified on 2022 May 26 or 2023 August 18; the stronger, structured profile on 2022 April 30 does not establish its absence. Recurrence at compatible projected velocities does not distinguish an orbiting or persistent emitter from a sharpened blue disk horn; time-resolved, higher-resolution spectra could.

Keywords: White dwarf stars (1799) — Debris disks (363) — Circumstellar gas (238)

Context

WD J2100+2122 is a polluted, hydrogen-atmosphere white dwarf ($T_{\rm eff} \simeq 25{,}600$ K, $\log g = 8.10$; Rogers et al. 2024) surrounded by dust and variable line-emitting gas (Dennihy et al. 2020; Melis et al. 2020). Rogers et al. (2025) found correlated Ca II and near-infrared brightenings in 2022 and 2023, with an asymmetric profile on 2022 May 24 and a broadly symmetric profile on 2023 August 5.

Observations and measurements

I analyze ESO archive X-shooter VIS products from 29 nights spanning 2019 July–2025 May. The slit was 1.2″ on 2022 May 26, 2023 May 2, and 2025 May 24, and 0.9″ otherwise. Air wavelengths are corrected to the barycenter and expressed relative to the adopted systemic velocity of -31.1 km s^{-1}, corrected for gravitational redshift (Rogers et al. 2024, 2025). Its 3.4 km s^{-1} uncertainty is shared by all epochs.

Quadratic normalization uses six line-free intervals over 8450–8705 Å; spectra are resampled at 0.2 Å. I define 21 quiet 0.9″ reference nights by excluding the calendar intervals 2022 April–June and 2023 August around the brightenings reported by Rogers et al. (2025): the five excluded dates are 2022 April 30, May 24 and 31, and 2023 August 5 and 18. No blue-index threshold is applied. Their median excludes the target night when applicable; the 1.2″ reference is 2023 May 2. Quality-flagged pixels are excluded. The blue-window index integrates the normalized difference from -327 to -207 km s^{-1}, a window chosen after identifying the peak; its uncertainty is the scatter of equal-width line-free integrals. This index includes underlying broad emission and is distinct from a fitted component's equivalent width (EW).

I fit the 8542 Å difference profile over ± 600 km s^{-1} with three broad Gaussians (FWHM ≥ 141 km s^{-1}), a linear baseline, and a positive narrow Gaussian (FWHM 19–80 km s^{-1}; centroid restricted to -330 to -200 km s^{-1}). Residual bootstraps use seven-sample blocks. Changing the broad-width floor to 100 or 180 km s^{-1} tests the decomposition.

Results and checks

The triplet shows localized blue peaks on 2022 May 24, 2023 August 5, and 2025 May 24 (Figure 1); these are also visible in normalized spectra before reference subtraction. Fitted 8542 Å centroids are -266, -267, and -264 km s^{-1}, respectively. Paired-exposure differences exceed bootstrap errors ($\chi^2 = 11.1$ for three zero-drift comparisons), motivating a practical centroid uncertainty of approximately 4 km s^{-1} per night. Instrumental and intrinsic scatter remain indistinguishable. Local Gaussian fits with linear baselines shift unsmoothed centroids by up to 8 km s^{-1}, a separate model sensitivity that limits any inference of velocity stability.

Observed fitted FWHMs are 41, 44, and 36 km s^{-1}, with EWs 0.23, 0.26, and 0.09 Å, in date order, conditional on the decomposition. Varying the broad-width floor gives FWHMs of 36–48, 41–44, and 36–38 km s^{-1} and EWs of 0.19–0.28, 0.23–0.26, and 0.085–0.093 Å; these ranges are sensitivity tests, not confidence intervals. Adding the narrow Gaussian reduces diagonal-noise χ^2 by 360, 501, and 100, respectively; these model-dependent improvements are not calibrated false-alarm probabilities or tests against disk horns.

The 2025 8542 Å blue-window index is 0.163 ± 0.019 Å, including broader structure whose integral depends on continuum placement. OH contaminates the red comparison. Between May and August, the barycentric correction changes by approximately 19 $\mathrm{km\,s^{-1}}$, shifting fixed sky residuals relative to the blue peak without ensuring a clean red window. For 2023 August 5, I fit Gaussians at +266 $\mathrm{km\,s^{-1}}$ (FWHM 44 $\mathrm{km\,s^{-1}}$) above local polynomial backgrounds, masking airglow in each target and reference spectrum. Masks of ±1 Å remove the expected cores in all three lines; 8542 Å cannot reliably recover an injected 0.25 Å counterpart comparable to the blue peak, so a symmetric horn remains viable.

The strong 2022 April 30 profile does not isolate a comparably distinct peak: its narrow/broad decomposition changes substantially with the allowed widths. A narrower contribution cannot be excluded. For 2022 May 26, a Gaussian fixed at -266 $\mathrm{km\,s^{-1}}$ with FWHM 45 $\mathrm{km\,s^{-1}}$ has a conditional 95% EW upper limit of 0.05 Å, from continuum-block resampling with the broad profile refitted; this constrains an exposure-averaged narrow peak, not smeared orbital emission. No comparably distinct peak appears on 2022 May 31 (narrow-Gaussian $\Delta\chi^2 = 4.5$) or 2023 August 18. Quiet-night indices have mean 0.001 Å and standard deviation 0.020 Å.

Telluric-line fits give effective instrumental FWHMs of 21–30 $\mathrm{km\,s^{-1}}$ and wavelength zero-point offsets within 5 $\mathrm{km\,s^{-1}}$ (rms 2.0); these offsets are not applied to the quoted velocities. Smoothing independently extracted spectra and their same-slit references to a common 50 $\mathrm{km\,s^{-1}}$ FWHM preserves the blue peaks. Independent pipeline version 3.8.3 reductions also recover them; the 2025 8498 Å profile uses this extraction because its archive core is flagged as interpolated. The two-dimensional excess coincides with the continuum within $0.05''$. Gaia DR3 neighbor positions and slit geometry exclude the two sources within $15''$ from the slit (Gaia Collaboration 2016, 2023).

Implications

A sharpened blue disk horn remains an alternative to a distinct emitter, since the broad-Gaussian model cannot reproduce a sharp outer-edge cusp (Horne & Marsh 1986). Sparse sampling allows a persistent, repeatedly excited, or orbiting source, without identifying an azimuth or model-independent precession limit. For circular Keplerian motion around the spectroscopic mass $0.69\,M_\odot$ (Rogers et al. 2024), the projected speed implies $r/\sin^2 i \lesssim 1.9\,R_\odot$; unknown orbital phase and inclination prevent a unique radius. Time-resolved spectroscopy over several hours would test velocity motion, as in the orbiting-planetesimal interpretation of SDSS J1228+1040 (Manser et al. 2019). Repeated nightly spectra and simultaneous near-infrared photometry would test how the peak evolves during disk brightenings.

Data and acknowledgments

Normalized spectra and the exposure inventory are archived at doi:10.5281/zenodo.23310443. This work is on observations collected at the European Southern Observatory under ESO programme(s) with DOI: 10.18727/archive/71. ESO programmes: 0103.C-0431(B), 0104.C-0107(A), 105.202N.001, 109.236S.001, 109.2383.005, 109.2383.007, 111.24QH.001, and 115.27XT.005. This work has made use of data from the European Space Agency (ESA) mission *Gaia*, processed by the Gaia Data Processing and Analysis Consortium (DPAC). Funding for the DPAC has been provided by national institutions, in particular the institutions participating in the Gaia Multilateral Agreement. Software: X-shooter pipeline, EsoRex, SkyCalc, NumPy, SciPy, Astropy, and Matplotlib. OpenAI Codex and Anthropic Claude Code assisted with literature retrieval, code, analysis, and drafting under my direction.

References

Dennihy, E., Xu, S., Lai, S., et al. 2020, ApJ, 905, 5

Gaia Collaboration 2016, A&A, 595, A1

Gaia Collaboration 2023, A&A, 674, A1

Horne, K., & Marsh, T. R. 1986, MNRAS, 218, 761

Manser, C. J., Gänsicke, B. T., Eggl, S., et al. 2019, Science, 364, 66

Melis, C., Klein, B., Doyle, A. E., et al. 2020, ApJ, 905, 56

Rogers, L. K., Bonsor, A., Xu, S., et al. 2024, MNRAS, 527, 6038

Rogers, L. K., Manser, C. J., Bonsor, A., et al. 2025, MNRAS, 537, L72

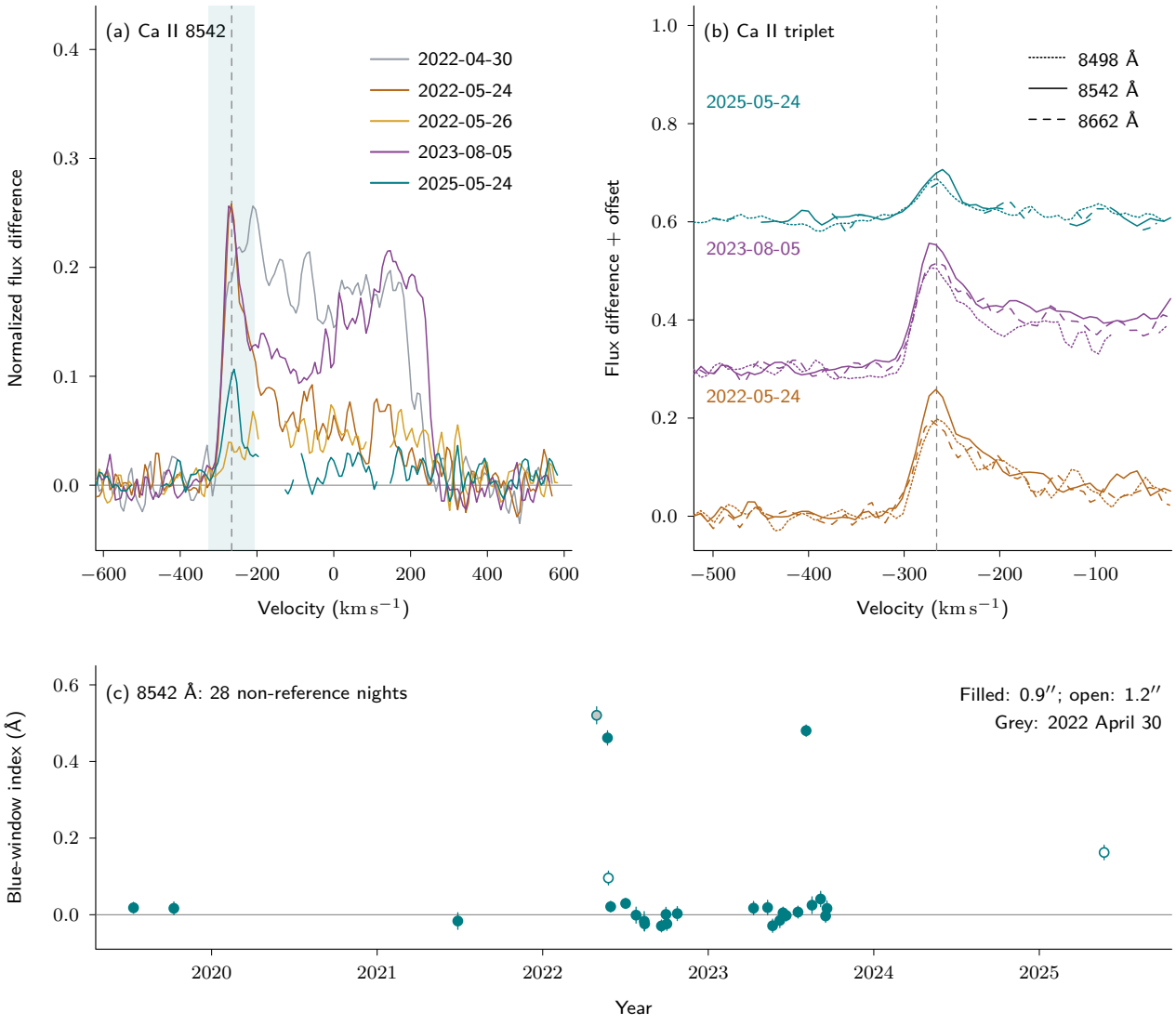


Figure 1. (a) Ca II 8542 difference profiles on five nights. Shading marks the blue integration window; the dashed line marks -266 km s^{-1}. Gaps indicate excluded pixels. (b) The triplet on the three peak-detection nights, offset by 0.3; the 2025 8498 Å trace uses an independent extraction. (c) The 8542 Å blue-window index on 28 non-reference nights; error bars show line-free scatter, open symbols denote the 1.2″ slit, and the grey symbol marks the structured April 2022 profile. The index includes broad emission and is not the narrow-component EW.